Khavda

The largest power plant site in the world, and the wires that cannot carry it

Field briefing · November 2026

Title background: the latest satellite frame, once `./khavda/make_frames.sh` has run.

{X}KDR

Six years in 28 seconds

One frame a quarter, 2020 to 2026, one second each

726 km²

of salt flat

30 GW

planned

14.8 GW

connected; 9 GW able to leave

Satellite frame not generated yet

The 28-frame quarterly animation, 2020 Q1 to 2026 Q4, one second per frame

`./khavda/make_frames.sh`

The same six years in six stills

Satellite frame not generated yet
2020 Q1: bare salt
`./khavda/make_frames.sh`

Satellite frame not generated yet
2021 Q4: the approach road
`./khavda/make_frames.sh`

Satellite frame not generated yet
2024 Q1: first arrays
`./khavda/make_frames.sh`

Satellite frame not generated yet
2025 Q3: KPS3 opens
`./khavda/make_frames.sh`

Satellite frame not generated yet
2026 Q1: the 5 GW year
`./khavda/make_frames.sh`

Satellite frame not generated yet
2026 Q4: latest
`./khavda/make_frames.sh`

Sentinel-2 cloud-free quarterly composites; contains modified Copernicus Sentinel data. Scene 48 × 27 km centred 24.117° N, 69.350° E.

Why here

- **Land nobody lives on.** State wasteland, unsurveyed, no one to buy out.
- **Sun second only to Ladakh.** 2,060 kWh per m² a year, 300 clear days.
- **Wind at 8 metres a second** off the Arabian Sea, strongest at night.
- **One kilometre from Pakistan.** Permits from the Border Security Force, for a site visit too.

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2020 Q1: the site before construction. White salt crust, the BSF road, India Bridge.

`./khavda/make_frames.sh`

The site before building, early 2020. Scale bar lower left.

The lease

72,600
hectares of state wasteland, one tract

40 years
ten to build, thirty to run

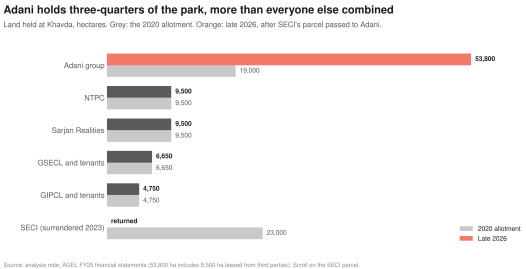
2 ha/MW
the allotment rule

Rs 15,000
per hectare a year; a fifth of private land nearby

The state removed the slowest step in Indian infrastructure, buying land, by leasing wasteland it already owned. The lease came with a clock: from the day the approach road was finished, December 2021, developers had three years to commission half their capacity and five to commission all of it.

Gujarat wasteland policy 2019; AGEL ESIA; ADB IEE for the SAEL plots.

Who holds the land



Satellite frame not generated yet

Latest frame with parcel outlines and hectares (needs `parcels.geojson` traced)

```
./khavda/make_frames.sh
```

Latest frame. Outlines and hectares from `parcels.geojson`; the orange trace is the measured built-up area.

What the plant is made of

1. Generate: panels and turbines
Bifacial panels on trackers, about 2,400 per MW. 5.2 MW turbines, 160 m rotor.

2. Convert and collect: inverters, 33 kV cables
Direct current to alternating; 1.36 MW of panels per MW of inverter. Cables underground, bedded in sand.

3. Pool: each developer's substations, 33 to 400 kV
Adani has 14, NTPC two, the others one or two each. Battery yards and voltage-support units sit beside them.

4. Export: Power Grid's stations, 400 to 765 kV
Three stations inside the park, KPS1 to KPS3. Two 765 kV lines out today; direct-current links from 2028.

What must exist around it

Roads raised 150 mm above flood level

Concrete mixed on site; Bhuj is 125 km

Water by tanker, rain tanks, desalting

Robots dry-clean panels most nights

Worker colonies with clinics on site

Own phone towers; one control room

Weather stations, fire station

Deep piles for salt, flood and quakes

Modules and turbines from Mundra

BSF permits; the border is 1 km away

Building on salt

45% salt
in the soil
Deep galvanised piles; cables bedded in sand

3 m
above sea level; floods in the monsoon
Roads raised 150 mm; inverters on plinths

Zone V
India's highest earthquake zone
Turbines on concrete rafts

46 °C
summers; 50 to 70 dust storms a year
Robots clean panels dry, three nights a week

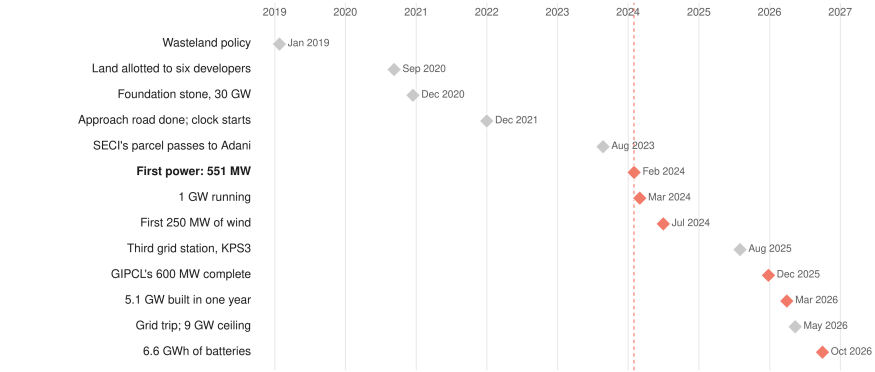
No water
the water table is hypersaline
Tankers, rain tanks, a desalination plant

Photo strip: a pile being driven, a raised road, a cleaning robot, a worker colony.
No ground photographs in the package yet; drop them in `figures/photos/` and replace this box.

The timeline

Five years to prepare the land, then 10 GW in under three

Milestones at Khavda, 2019 to 2026. Orange: building. The dashed line marks the first power.

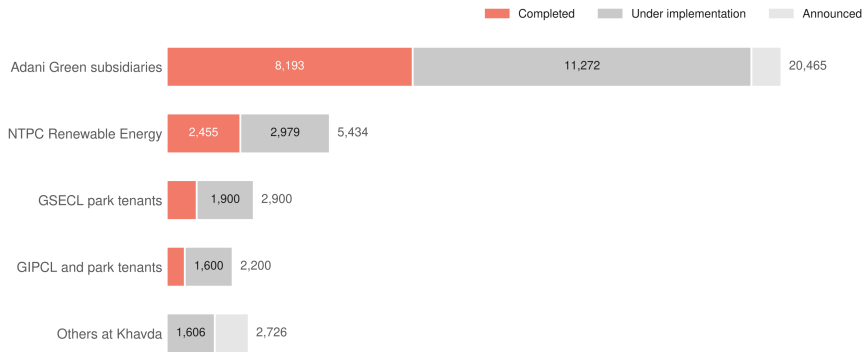


Source: company statements, the WRPC report, Scroll's RTI records. Data: timeline_milestones.csv

What the CMIE capex data shows

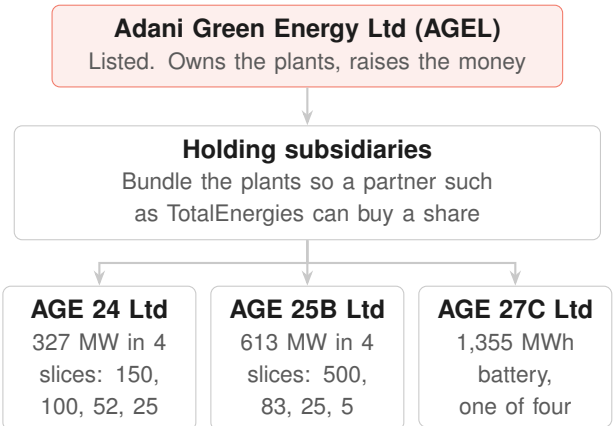
Adani has built two-thirds of what is complete at Khavda, and holds most of the pipeline

Generation capacity in MW from the CMIE CapEx extract: 12,248 completed, 19,357 under implementation.



Source: CMIE CapEx, Nov 2026; 157 projects, 8 park-level umbrella records removed; generation only. Excludes 16,000 MW of multi-site announcements by Tata Power and Serentica.

How Adani builds



Three of 24 numbered subsidiaries in the CMIE sheet. Each

The group supplies every part

Adani New Industries: panels and turbines, made at Mundra

Adani Ports: brings in the kit

Ambuja and ACC: the cement

Adani Energy Solutions: builds the lines out, both DC links

Adani Infra: the contractor, even on NHPC's 600 MW

Most of each rupee stays in the group

Everyone else

Tenant	Park, status	MW	Rs cr/MW
SAEL Solar P4 and P5	GSECL, completed	800	6.5
KPI Green Energy	GSECL, completed	200	7.8
NLC India	GSECL, building	600	5.6
SJVN Green (three plants)	GSECL, building	600	4.3 to 5.5
JSW Renew Energy Ten	GSECL, building	300	–
NHPC	GSECL, building	200	4.6
Enren-II Energy	GSECL, building	200	6.5
GIPCL (own plant)	GIPCL, completed	600	6.5
SJVN and SJVN Green	GIPCL, building	700	5.4 to 6.5
SAEL Solar P6	GIPCL, building	400	3.25
Coal India	GIPCL, building	300	6.5
NHPC	GIPCL, building	200	4.1

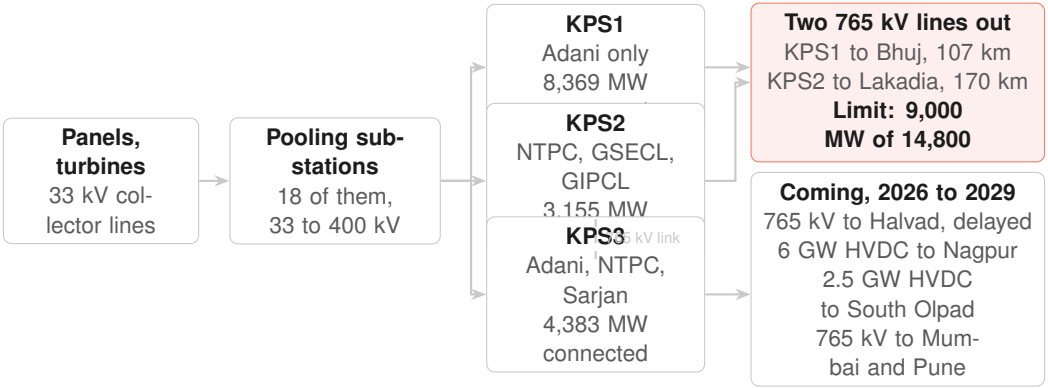
Two models side by side. Adani and NTPC build their own plants. GSECL and GIPCL are landlords: they build substations and roads, then lease plots at 1.85 ha per MW to whoever wins a GUVNL auction.

The tenants are the comparison group. Same land, same rent, same technology, but bought at auction.

CMIE records some at Rs 3.25 to 5.5 crore per MW against Adani's 6.5. Worth asking why on site.

CMIE CapEx extract, Nov 2026. Costs are filed at announcement. NTPC not shown: 2.5 GW done, builds its own.

Getting the power out

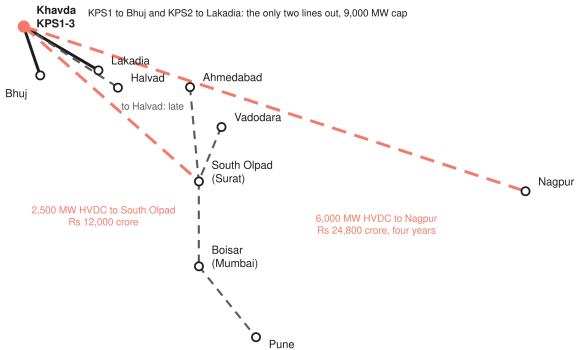


Panels take a year; lines take three to five. The park outran its wires, so the operator caps it at 9,000 MW. Source: WRPC grid disturbance report, May 2026, connected MW by station.

The corridors out of Kutch

Two lines leave the park today; Rs 60,000 crore of lines are still being built

Corridors out of Kutch. Solid: in service. Dashed: under construction. Orange: the two direct-current links.

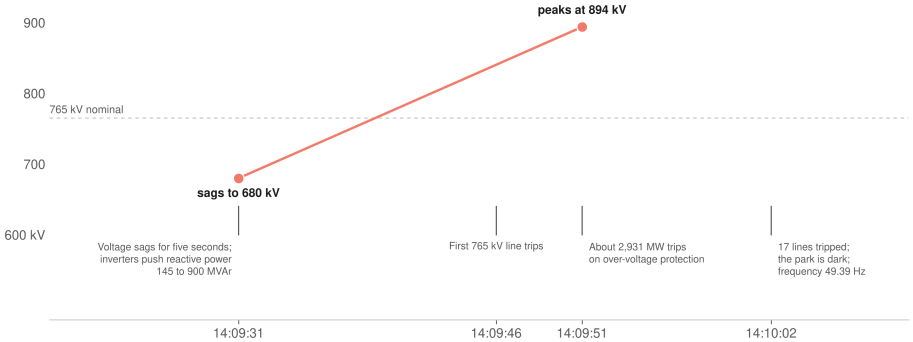


Schematic; positions approximate. Source: CTUIL minutes, CMIE transmission projects (transmission_projects_cmie.csv).

13 May 2026: dark in 17 seconds

Dark in 17 seconds: the voltage overshot and seventeen lines tripped

The 765 kV system around Khavda on 13 May 2026. Orange: the recorded voltage. The park had been at its 8,963 MW ceiling since 14:00.

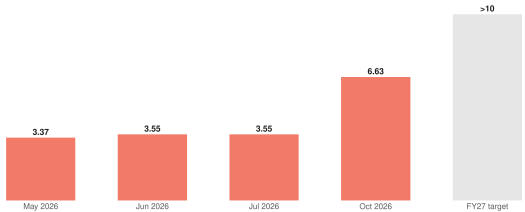


Source: WRPC grid disturbance report on the Khavda event; the four earlier steps (fault near Indore 14:02:39, oscillation 14:03:46, voltage dips from 14:05:40) are in the note. Data: blackout_13may2026.csv

Batteries as the stopgap

Adani doubled its batteries at Khavda in four months, to half of India's total

Battery storage operating at Khavda, GWh. Light grey: the FY27 target of more than 10 GWh, Rs 15,000 crore.



Source: WRPC report (May 2026), AGEL statements (Jun, Jul, Oct 2026), AGEL FY27 guidance. Data: battery_gwh.csv

When the line is full at noon, power is lost. Batteries store it and sell it at 7 pm.

6.6 GWh

in October 2026, half of India's total

Rs 15,000 cr

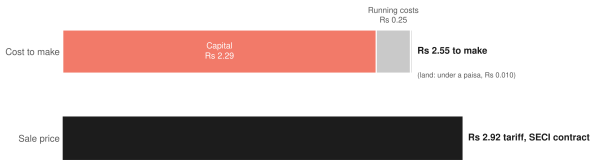
to pass 10 GWh in FY27

Curtailment could cost 5 to 7% of group EBITDA until the lines open. On the satellite image the battery yards are the dense rectangles beside the switchyards.

The economics

A unit costs about Rs 2.5 to make and sells for Rs 2.92

Rupees per kWh. Capital: Rs 6.5 crore per MW at 33% capacity factor, 9% over 25 years. Land rent: Rs 30,000 per MW a year.



Source: CMIE cost convention; AGEL SECI manufacturing-linked tariff; analysis note. The Rs 6.5 crore is a filed convention, not an outturn.

Who buys the power

25-year contracts with SECI, NTPC and GUVNL.

Corporates: Google (Oct 2024), Asahi India Glass (Jan 2026).

The evening exchange market, for battery power.

When finished

81 to 87 billion units a year: about 4% of what India uses.

Adani, in four points

1. Got the private cap of land, 19,000 ha, in 2020, and the 23,000 ha SECI parcel in 2023 without a tender.
2. Faced a US indictment in November 2024 over the SECI contracts; dropped and settled in 2026 with no admission.
3. Built 5.1 GW at Khavda in the year that straddled it.
4. Its own transmission arm builds the lines out, including both direct-current links.

Neither the share-price shock of 2023 nor the indictment of 2024 slowed the build. The slow lines cost one Adani company, AGEL, lost power, and pay another, Adani Energy Solutions, to build them.

The plant in one page

Inside the park: fast. One owner, flat land, factory kit, 5 GW a year.

Outside: slow. Farmland, forests, three to five years a line.

Between: batteries. 9 GW cap, lost noon power, 6.6 GWh of storage.

The question for the visit: when the lines open in 2027 to 2029, does the thin margin hold, and does the landlord-tenant model prove cheaper than Adani's own?

Satellite frame not generated yet

Latest frame, annotated: the three KPS stations and the two lines out

`./khavda/make_frames.sh`

Latest frame. Mark KPS1 to KPS3 and the two corridors as polygons in `parcels.geojson`.

What the frames measure

Satellite frame not generated yet

Built-up hectares per quarter, measured from the frames, against the milestones. Appears after `make_frames.sh` then `make_figures.py`.

```
./khavda/make_frames.sh
```

Sources